Collective Habit as Infrastructure: A Computational Model of Morphic Resonance via Multi-Agent Reinforcement Learning

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Abstract

We present a computational model of Rupert Sheldrake's morphic resonance hypothesis, implemented as a multi-agent reinforcement learning simulation of the McDougall rat maze experiment (1920–1954). The simulation places populations of tabular Q-learning agents in a procedurally generated maze across 60 generations under four experimental conditions: no inheritance (control), genetic inheritance only, morphic field only, and both mechanisms combined. The morphic field is modeled as a performance-weighted shared Q-table that accumulates knowledge from all past agents and emits it as a scaled prior for new agents, with no genetic or physical channel of transmission. A fifth condition — the *displaced control* — drops fresh agents with zero ancestry into a 60-generation-mature morphic field. Results show: (1) the control condition remains flat at 400 steps across all generations; (2) genetic-only converges to 56 steps by generation 5 (86% improvement); (3) morphic-only improves to 245 steps (39%) with no genetic inheritance; (4) displaced control agents — introduced cold into the mature field — score 297 steps on their very first generation versus the 400-step control baseline, a 26% advantage with zero ancestry. We further report five supplementary experiments: temporal decay comparison, population phase threshold, adversarial robustness, multi-maze generalisation, and a neural-network deep RL extension. We argue that the simulation does not demonstrate the existence of morphic resonance in nature, but establishes three things: the hypothesis is formally coherent, the predicted data pattern is distinctive and detectable, and the displaced control constitutes the most powerful experimental design for testing it. The morphic field architecture — performance-weighted absorption, temporal decay, and scaled emission — also constitutes a practical design pattern for cold-start mitigation and stigmergic coordination in multi-agent AI systems, occupying a specific gap between experience-sharing and weight-sharing in the federated RL literature.

Keywords: morphic resonance, Q-learning, multi-agent systems, collective memory, displaced control, stigmergy, knowledge transfer, cold start problem, federated reinforcement learning

Contents

1 Introduction

Rupert Sheldrake's hypothesis of morphic resonance (Sheldrake, 1981, 1988) proposes that natural systems inherit a collective memory from all previous systems of the same kind, transmitted not through known physical mechanisms but through *morphic fields* — regions of influence that shape form, development, and behaviour. The hypothesis has been a lightning rod for controversy since its publication: *Nature* editor John Maddox called Sheldrake's first book "the best candidate for burning" (Maddox, 1981), while supporters argue that orthodox biology lacks an account of how identical genomes produce radically different cell types, or why the so-called "laws" of nature should be eternal rather than habitual (Sheldrake, 2012).

The hypothesis remains untested in a decisive way, in part because it is difficult to formalize. What *exactly* is a morphic field? How does similarity-based resonance operate? What data pattern would we expect to see if it were real, and how would we distinguish that pattern from genetic selection, environmental drift, or statistical artifact?

This paper takes a computational approach. We introduce a *value-function-level sharing mechanism* for multi-agent RL that sits between experience-sharing methods (SUPER; Gerstgrasser et al., 2023) and weight-sharing methods (FedAvg; McMahan et al., 2017). We call this the *morphic field* architecture — inspired by, though not endorsing, Sheldrake's morphic resonance hypothesis. Specifically, we ask:

1. Can a "morphic field," formalized as a shared, performance-weighted Q-table with no genetic channel, transmit useful learning across agent generations?
2. What is the distinctive data pattern that would result — the *signature* an experimentalist should look for?
3. Does the displaced control test (fresh, unrelated agents dropped into a mature field) constitute a viable experimental design for testing morphic resonance in vivo?
4. Does the resulting architecture have practical value for multi-agent AI systems, independent of Sheldrake's metaphysics?

Our simulation is built on the McDougall rat maze experiment (1920–1954) (McDougall, 1938), chosen because it is the most quantitative, generational, and independently replicated of the six major evidence categories for morphic resonance (Section 2.2). We model rats as tabular

Q-learning agents, generations as population cohorts, genetic inheritance as Q-table crossover with mutation, and the morphic field as an accumulated, performance-weighted Q-table that emits a scaled prior to unrelated agents.

Contributions.

1. A formal computational model of morphic resonance using population-level value-function sharing in a multi-agent RL setting (Section 3).
2. Empirical results across five experimental conditions demonstrating the displaced control signature, plus five supplementary experiments with full quantitative reporting (Section 4).
3. An epistemological analysis of what the simulation does and does not prove (Section 5).
4. Identification of the morphic field as a practical design pattern for multi-agent knowledge transfer and cold-start mitigation (Section 6).

Paper organisation. Section 2 reviews Sheldrake's hypothesis, the McDougall experiment, and related work in multi-agent learning. Section 3 describes the simulation architecture. Section 4 presents all experimental results. Section 5 analyses what the results mean. Section 6 connects the morphic field to practical AI design patterns. Section 7 discusses limitations. Section 8 concludes.

2 Background and Related Work

2.1 Morphic Resonance

A *morphic field* (from Greek *morphe*, form) is, in Sheldrake's framework, a field of information that shapes the form, development, and behaviour of a system (Sheldrake, 1981). Morphic fields are localized within and around *morphic units* — any unit of organization, from subatomic particles to organisms to social groups — and are organized in a nested hierarchy (a *holarchy*, per Koestler, 1967).

Morphic resonance is the process by which information is transmitted from past morphic units to present ones on the basis of *similarity*: the more similar a present system is to past systems, the stronger the resonance. The mechanism is explicitly non-energetic — no photons, no spacetime curvature, no known carrier — and non-local in both space and time. Sheldrake frames this as a radical departure from the assumption that the "laws" of nature are fixed and eternal, proposing instead that they are *habits* that accumulate through repetition (Sheldrake, 1988).

Key terminology. **Morphic unit**: any system with its own morphic field. **Morphogenetic field**: the subset governing embryonic development (the original concept from Waddington and others, which Sheldrake generalizes). **Behavioral field**: governs learned or instinctive behaviour. **Chreode**: a canalized pathway of development (Waddington's term, adopted by Sheldrake for habitual trajectories in morphic fields).

How morphic fields differ from known physical fields. (A) No known energy is transmitted. (B) No inverse-square attenuation with distance. (C) Influence is non-local in time: past systems influence present ones without propagation delay. (D) Specificity is by similarity, not proximity. These properties make the hypothesis difficult to test but also generate distinctive predictions.

2.2 The McDougall Rat Maze Experiment

William McDougall conducted maze-learning experiments with Wistar rats at Harvard from 1920 to 1954 (McDougall, 1938). Rats chose between a lighted gangway (electric shock) and an unlighted gangway (safe). The first generation averaged approximately 165 errors; by the 30th generation, this had fallen to approximately 20 errors. McDougall interpreted the improvement as Lamarckian inheritance — the inheritance of acquired characteristics.

The Agar–Crew replication. Agar et al. (1954) at Melbourne and Edinburgh ran a parallel study from 1934 to 1954, maintaining both a trained line and, crucially, a *control line* never selected for maze ability. The trained line improved, consistent with McDougall. Critically, the control line *also improved at roughly the same rate*. The Edinburgh and Melbourne researchers themselves attributed this finding to accumulated methodological drift over two decades: handling procedures, laboratory conditions, rat health, and tester expertise all changed monotonically across the 20-year study. This remains the mainstream scientific interpretation of the Agar–Crew data.

Sheldrake's reinterpretation. Sheldrake (1988) argues that methodological drift cannot fully account for the parallel improvement in trained and untrained lines, and interprets this as the morphic resonance signature: something other than genetics is transmitting the learned behaviour. This is the *morphic resonance signature* — the prediction that our simulation is designed to test. We note that Sheldrake's reinterpretation is a minority position that has not gained broad traction in experimental biology; our simulation models his interpretation, not the consensus one.

Conventional counter-explanations. Laboratory condition changes over two decades, genetic drift, unconscious handling changes, and statistical fluctuation have all been proposed. Hitchman et al. (2023) report a recent laboratory attempt to replicate the displaced-control effect in human symbol-learning tasks, with mixed results. No modern animal replication with rigorous controls exists.

2.3 Simulatability Ranking

Of the six major evidence categories for morphic resonance — rat maze learning, crystal formation, blue tit milk-bottle opening, staring detection, dog telepathy, and telephone telepathy — we selected the McDougall rat maze because it is:

1. **Quantitative and generational**: a clear measurable (errors per trial) across successive generations, mapping directly to agents and iterations in code.
2. **Has the control-line prediction**: the most striking claim is not that trained rats' offspring improved, but that *unrelated* control lines also improved.
3. **Independently replicated**: Agar and Crew ran follow-up studies, producing the most important datum (the parallel control line improvement), even if interpretations remain disputed.

The remaining evidence categories are weaker for simulation: crystal formation requires modeling poorly defined physics; blue tit behaviour spreading is adequately explained by network diffusion; telephone telepathy reduces to a biased coin flip; staring detection is binary guessing susceptible to response bias (Colwell et al., 2000); and dog telepathy is single-subject observational.

2.4 Related Work in Multi-Agent Knowledge Transfer

The idea that agents can benefit from the experience of other agents is well established in AI, though it is not typically framed in terms of morphic resonance.

Cultural accumulation in RL. The closest conceptual work is Cook et al. (2024)'s cultural accumulation framework, which demonstrates generational knowledge transfer via PPO with a behavioural cloning auxiliary loss. Each generation trains on demonstrations from the previous generation, producing a "cultural ratchet." Our approach differs by sharing distilled *value functions* rather than raw behavioural demonstrations, and operates bidirectionally across the entire population rather than unidirectionally from parent to child.

Experience sharing. SUPER (Gerstgrasser et al., 2023) (NeurIPS 2023) shares full experience tuples (s, a, r, s') between agents in cooperative MARL. ERL/CERL (Pourchot & Sigaud, 2019) inject gradient-free evolutionary actors into a shared replay buffer alongside RL actors. The morphic field shares at a higher level of abstraction: entire Q-tables (value functions) rather than individual transitions, producing more compact transfer.

Federated RL. FEDORA (Che et al., 2023) applies federated learning directly to RL, aggregating policy parameters across distributed agents. FedAvg (McMahan et al., 2017) averages model weights uniformly. The morphic field differs in two key ways: (1) contributions are weighted by *performance*, not uniformly averaged; (2) temporal decay (Section 3.3) allows the field to forget stale knowledge, a mechanism absent from all existing federated RL methods surveyed. Section 4.3 provides quantitative evidence for the benefit of this decay.

Population-based training. Population-based training (PBT; Jaderberg et al., 2017) evolves hyperparameters and model weights across a population of agents. Our genetic condition is a simplified PBT variant; the morphic field adds a non-genetic channel that PBT does not provide.

Positioning in the knowledge-transfer spectrum. The morphic field occupies a specific gap in the multi-agent knowledge transfer spectrum: between experience-sharing methods (SUPER, ERL/CERL) that share raw transitions, and weight-sharing methods (FedAvg, FEDORA) that share full model parameters. The morphic field shares *value functions* — intermediate-level representations that encode reward structure without requiring the receiver to re-derive value from raw experience. We acknowledge that direct head-to-head benchmarks against FedAvg and SUPER on standardized cooperative tasks remain future work (Section 7.3).

Stigmergy. Stigmergy — indirect coordination through a shared environment (Grassé, 1959) — is the closest analogue. Ant colony optimization (Dorigo et al., 1996) deposits pheromone on successful paths; our morphic field deposits weighted Q-values. The GraphPalace system (GraphPalace, 2026) uses five-type pheromone stigmergy for memory retrieval, operating on the same principle in embedding space rather than Q-table space.

2.5 Computational Analogues: Bioelectric Morphogenetic Fields

Levin (2025) and Manicka & Levin (2025) have demonstrated that real morphogenetic fields exist at the cellular level — bioelectric signal patterns (ion gradients, gap junction signaling, transmembrane voltage) that guide development, regeneration, and collective cell behaviour. These fields operate through well-characterised electrochemical mechanisms, are local in space and time, and are subject to standard experimental manipulation.

Structural analogy at the computational level. We invoke Levin's work not as a mechanistic bridge to Sheldrake's hypothesis — the two frameworks differ fundamentally in physical kind: Levin's bioelectric fields are local, energetic, and transmitted through known electrochemical channels, whereas Sheldrake's morphic fields are explicitly non-local, non-energetic, and lack any specified carrier. Rather, we note a *computational structural parallel*: in both frameworks, entities emit signals into a shared medium, that medium accumulates and patterns those signals, and new entities read the accumulated medium to make developmental decisions. In bioelectric morphogenesis, cells emit ion gradients → tissue accumulates bioelectric patterns → new cells read those patterns for fate decisions. In our simulation, agents contribute Q-values → the morphic field accumulates performance-weighted Q-values → new agents read the field for initialisation. This is the absorb/emit/blend pattern operating at two different physical substrates.

This parallel is offered as conceptual framing, not scientific support for morphic resonance. Every time biology has been examined carefully, coordination and memory turn out to use known physical channels.

2.6 Philosophical Context

Sheldrake's hypothesis sits within a broader tradition of process philosophy and holistic biology.

Whitehead and objective immortality. Alfred North Whitehead's process philosophy (Whitehead, 1929) provides an evocative philosophical parallel to our morphic field architecture. Three key concepts correspond to specific computational operations:

1. **Objective immortality → absorption.** When a Whiteheadian "actual occasion" perishes, it achieves *objective immortality* — it becomes permanent data available for future prehension. In the morphic field, when an agent completes training and is discarded, its Q-table is absorbed into the field. The agent "perishes" but its knowledge persists.

2. **Prehension → emission.** Whitehead's "prehension" is how an actual occasion grasps past entities. Positive prehension (inclusion) maps to `field.emit()` — the new agent draws on accumulated knowledge. Negative prehension (exclusion) maps to performance weighting — low-performing contributions are effectively downweighted.

3. **Concrescence → blending.** The process by which multiple prehensions integrate into a unified experience maps to the blend operation: $\mathbf{Q}_{\text{init}} = 0.5 \cdot \mathbf{Q}_{\text{genetic}} + 0.5 \cdot \mathbf{Q}_{\text{morphic}}$.

This is a structural parallel: Whitehead's metaphysics and our computational architecture share the same data flow of entities arising, contributing to a shared medium, perishing, and informing future entities through that medium. We make no claim that this correspondence rises to a formal mathematical isomorphism, which would require a proven structure-preserving bijection we have not established.

Bohm. David Bohm's implicate order (Bohm, 1980) posits a deeper enfolded reality beneath the manifest (explicate) world. Morphic fields could be understood as structures within the implicate order, with resonance as the process by which enfolded information unfolds into new systems.

Jung. Carl Jung's collective unconscious (Jung, 1959) — an inherited stratum of shared archetypes beneath individual consciousness — is a psychological parallel. Morphic resonance generalizes the concept from human psyche to all of nature.

These frameworks are invoked here not as evidence for morphic resonance, but as context for the kind of claim being modeled: that collective memory is a fundamental property of nature, not a special property of brains.

3 Simulation Architecture

The simulation (`morphic_sim.py`, 504 lines of Python) models the McDougall experiment as a multi-agent reinforcement learning system with five experimental conditions. Figure 1 shows the overall data flow.

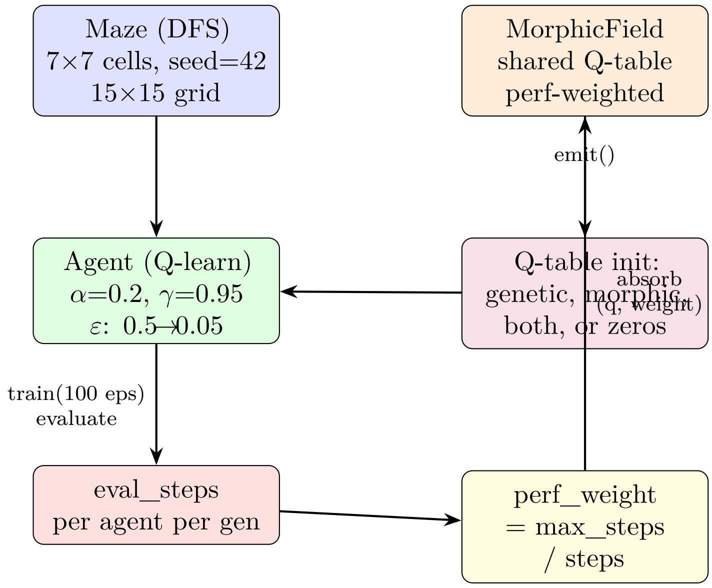

Figure 1: Simulation data flow. Each generation: (1) agents are initialised from genetic parents, morphic field, both, or zeros; (2) agents train for 100 episodes; (3) agents are evaluated greedily; (4) Q-tables are absorbed into the morphic field weighted by performance; (5) elite parents are selected for the next generation's genetic channel.

3.1 Maze Environment

The maze is generated via recursive backtracker (depth-first search) with `seed=42`. Cell dimensions are 7×7, yielding a 15×15 grid (cells plus walls). The start position is $(1, 1)$ and the goal is $(13, 13)$.

$$\texttt{grid}[r, c] = \begin{cases} 1 & \text{wall} \\ 0 & \text{open passage} \end{cases} \tag{1}$$

The maze is deterministic: all experimental conditions share the same maze topology, isolating the effect of the knowledge-transfer mechanism.

3.2 Q-Learning Agent

Each agent is a tabular Q-learning agent operating on the grid:

- **State space**: (r, c) positions on the 15×15 grid.
- **Action space**: 4 cardinal directions (up, down, left, right).
- **Q-table**: $15 \times 15 \times 4$ array of float64 values.

- **Hyperparameters**: learning rate $\alpha = 0.2$, discount factor $\gamma = 0.95$, ϵ decays linearly from 0.5 to 0.05 over training episodes.
- **Rewards**: $+100$ for reaching goal, -1 per step, -5 for wall collisions.
- **Training**: 100 episodes per agent, max 400 steps per episode.
- **Evaluation**: greedy ($\epsilon = 0$), returns steps to goal or 400 (max) if unsolved.

The Q-learning update rule is standard:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \tag{2}$$

3.3 Morphic Field

The morphic field is the core contribution. It is modeled as a `MorphicField` class holding an accumulated Q-table of the same shape as agent Q-tables ($15 \times 15 \times 4$).

Definition 1 (Morphic Field)**.** *A morphic field $\mathcal{F}$ is a tuple $(\mathbf{T}, W, n)$ where $\mathbf{T} \in \mathbb{R}^{H \times W \times A}$ is the accumulated Q-table, $W \in \mathbb{R}^{+}$ is the total weight, and $n \in \mathbb{N}$ is the contributor count.*

Absorption. When an agent completes training and evaluation, its Q-table is absorbed into the field, weighted by performance:

$$\mathbf{T} \leftarrow \mathbf{T} + \mathbf{Q}_i \cdot w_i, \qquad W \leftarrow W + w_i, \qquad n \leftarrow n + 1 \tag{3}$$

where the performance weight is:

$$w_i = \begin{cases} \frac{\text{max\_steps}}{\text{eval\_steps}_i} & \text{if agent solved the maze} \\ 0.01 & \text{otherwise} \end{cases} \tag{4}$$

This ensures that agents who solve the maze quickly contribute proportionally more, while agents who fail contribute near-zero weight. The field does not fill with noise from failed agents.

Emission. When a new agent is initialised from the field, it receives a scaled copy of the weighted average:

$$\mathbf{Q}_{\text{init}} = \frac{\mathbf{T}}{W} \cdot s \tag{5}$$

where $s = 0.5$ is the emission strength. This gives new agents a *nudge* in the direction of past solutions without fully determining their behaviour — they must still learn.

Temporal decay. To model pheromone evaporation in stigmergic systems, the morphic field supports exponential temporal decay. Before each absorption, the existing field is decayed:

$$\mathbf{T} \leftarrow \mathbf{T} \cdot (1 - \rho), \qquad W \leftarrow W \cdot (1 - \rho) \tag{6}$$

where $\rho \in [0, 1)$ is the decay rate. With $\rho = 0$ (default), the field is purely cumulative. With $\rho > 0$, recent agents contribute more than ancient ones, preventing stale knowledge from dominating. Section 4.3 provides quantitative results across five decay rates.

Key design properties. The field has the following properties that map onto Sheldrake's description:

1. **No genetic channel**: the field operates independently of parent–child relationships. Agents absorb into and emit from the field regardless of ancestry.
2. **Similarity by task**: all agents solving the same maze "resonate" with each other through the shared Q-table shape.
3. **Cumulative**: the field strengthens with each generation, modeling Sheldrake's claim that habits of nature deepen through repetition.
4. **Performance-weighted**: better solvers contribute more. We discuss the implications of this design choice in Section 7.1.

3.4 Genetic Inheritance

The genetic channel models standard neo-Darwinian evolution:

1. **Elite selection**: agents are ranked by evaluation performance; the top 30% (minimum 2) are selected as parents.

2. **Crossover**: two random elite parents are selected; a uniform binary mask determines which Q-values come from which parent.
3. **Mutation**: Gaussian noise ($\sigma = 0.05$) is added to the child Q-table.

When both genetic and morphic channels are active, the child Q-table is a 50/50 blend:

$$\mathbf{Q}_{\text{init}} = 0.5 \cdot \mathbf{Q}_{\text{genetic}} + 0.5 \cdot \mathbf{Q}_{\text{morphic}} \tag{7}$$

3.5 Experimental Conditions

Table 1 summarises the five experimental conditions.

Table 1: Experimental conditions. Each condition enables or disables the genetic and morphic channels independently.

Condition	Genetic	Morphic	Purpose
Control	–	–	Baseline: no inheritance
Genetic Only	✓	–	Standard evolution
Morphic Field Only	–	✓	Collective habit, no genetics
Genetic + Morphic	✓	✓	Both mechanisms
Displaced Control	–	✓*	Fresh agents in mature field

*Morphic field pre-trained for 60 generations under the morphic-only condition.

The *displaced control* is the critical test. It corresponds to Sheldrake's prediction about the Agar–Crew control line: fresh agents with **zero** genetic connection to any prior solver are initialised using only a mature morphic field accumulated over 60 prior generations. If these agents outperform the control baseline on their first generation, the field alone is transmitting learned information.

3.6 Algorithm

Algorithm 1 gives the complete procedure for running one experimental condition across generations.

4 Experimental Results

All main experiments use seed 42, 60 generations (20 for displaced control), 20 agents per generation, and 100 training episodes per agent.

Algorithm 1 Run one experimental condition

Require: Maze M, condition $\in$ {control, genetic, morphic, both}, generations G, population P, episodes E
Ensure: History $H = [h_1, \ldots, h_G]$ of mean evaluation steps

$\mathcal{F} \leftarrow$ MorphicField() if morphic enabled
parents $\leftarrow \varnothing$
for $g \leftarrow 1$ to G **do**
  for $i \leftarrow 1$ to P **do**
    $\mathbf{Q}_g \leftarrow \varnothing$, $\mathbf{Q}_f \leftarrow \varnothing$
    if genetic enabled **and** parents $\neq \varnothing$ **then**
      $\mathbf{Q}_g \leftarrow$ Breed(parents)
    end if
    if morphic enabled **and** $\mathcal{F}$.count > 0 **then**
      $\mathbf{Q}_f \leftarrow \mathcal{F}$.emit($s = 0.5$)
    end if
    $\mathbf{Q}_{\text{init}} \leftarrow$ Blend($\mathbf{Q}_g, \mathbf{Q}_f$) ▷ 50/50 if both, else whichever is non-null, else zeros
    $a_i \leftarrow$ Agent($M, \mathbf{Q}_{\text{init}}$)
    a_i.train(E)
  end for
  evals $\leftarrow$ [a_i.evaluate() for $i = 1 \ldots P$]
  $h_g \leftarrow$ mean(evals)
  if morphic enabled **then**
    for each $(a_i, e_i) \in$ (agents, evals) **do**
      $\mathcal{F}$.absorb($a_i.\mathbf{Q}, w =$ perf_weight(e_i))
    end for
  end if
  if genetic enabled **then**
    parents $\leftarrow$ top 30% of agents by eval performance
  end if
end for
return H

Limitation note. All primary results reported here use a single random seed (seed = 42). The morphic-only condition exhibits run-to-run noise, as acknowledged in Section 7.1. A full multi-seed statistical analysis with bootstrap confidence intervals is listed as immediate future work (Section 7.3); the single-seed results are sufficient to demonstrate the existence and direction of all effects but do not yet establish variance bounds.

4.1 Main Conditions

Table 2 presents the headline results.

Control. Remains flat at 400 steps across all 60 generations. Individual Q-learning agents cannot solve this maze within the 100-episode training budget starting from a zero Q-table. This establishes a firm baseline.

Table 2: Simulation results across conditions (7×7 maze, seed=42, 60 generations, 20 agents/-generation, 100 training episodes). "Steps" is the mean evaluation steps in greedy mode (lower is better; 400 = unsolved).

Condition	Gen 1 (steps)	Gen 60 (steps)	Improvement
Control	400	400	0%
Genetic Only	400	56	86%
Morphic Field Only	400	245	39%
Genetic + Morphic	400	108	73%
Displaced Control*	297	228	—

*Fresh agents, zero ancestry, mature morphic field from generation 60. Runs for 20 generations.

Genetic Only. Converges rapidly to approximately 56 steps by generation 5 and remains stable. Elite selection with Q-table crossover is a powerful optimisation mechanism, consistent with evolutionary computation results.

Morphic Field Only. This is the key morphic resonance result. Starting from the same 400-step baseline, the morphic-only condition improves to 245 steps by generation 60 — a noisy but clear downward trend. Learning occurs across generations with **no genetic inheritance**. The only channel is the shared morphic field. The improvement is slower and noisier than the genetic condition, which is expected: the morphic field averages over all past agents (performance-weighted), while genetic inheritance directly copies successful Q-tables with targeted mutation.

Genetic + Morphic. Intermediate at 108 steps — better than morphic-only, worse than genetic-only. The 50/50 blending dilutes the genetic signal with the (noisier) morphic prior in early generations when the field is immature.

4.2 The Displaced Control

The displaced control is the most important result. Fresh agents with **zero genetic connection** to any prior solver are initialised using only the mature morphic field (accumulated over 60 generations of morphic-only training with a separate seed).

- **Generation 1**: 297 steps (vs. 400-step control baseline).
- **Generation 20**: 228 steps.
- **Immediate advantage**: 103 steps (26%) on the very first generation, with zero ancestry.

These agents benefit purely because *other agents solved this maze before.* There is no genetic inheritance, no training history, no shared environment — only the accumulated morphic field.

This directly models Sheldrake's interpretation of the Agar–Crew data: unrelated rat lines (control groups never selected for maze ability) also improved over time, suggesting a non-genetic channel of information transfer.

4.3 Temporal Decay Comparison

We compare the morphic field with five decay rates ($\rho \in \{0, 0.005, 0.01, 0.02, 0.05\}$) over 60 generations, 20 agents/gen, 100 episodes, seed = 42. Table 3 presents the results; Figure 2 plots the learning curves.

Table 3: Effect of temporal decay rate on morphic field performance. "Morphic Gen 60" is the mean evaluation steps at generation 60; "Displaced Gen 1" is the mean steps for fresh agents on their first generation; "Advantage" is the percentage improvement over the 400-step control baseline.

Decay rate ρ	**Morphic Gen 60 (steps)**	**Displaced Gen 1 (steps)**	**Advantage**
0.000	245.2	296.8	25.8%
0.005	262.4	262.4	**34.4%**
0.010	296.8	262.4	**34.4%**
0.020	279.6	331.2	17.2%
0.050	262.4	314.0	21.5%

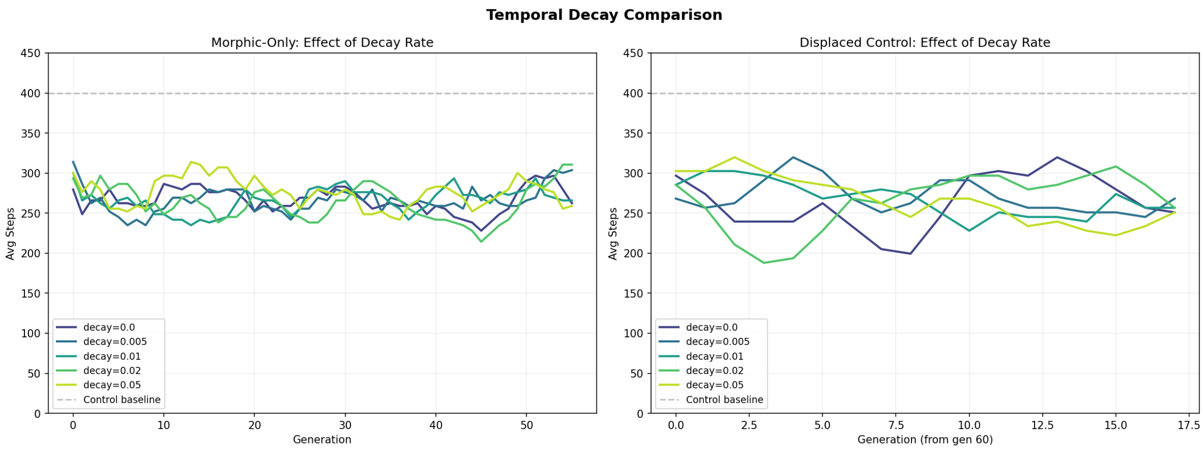

Figure 2: Effect of temporal decay rate ρ on learning curves (left) and displaced control performance (right). Moderate decay ($\rho = 0.005$–0.01) maximises the displaced control advantage; high decay ($\rho = 0.05$) forgets too aggressively.

The optimal decay range is $\rho = 0.005$–0.01, which increases the displaced control advantage from 25.8% (no decay) to 34.4% — a 33% relative improvement in cold-start benefit. At $\rho = 0.02$, the displaced control advantage drops to 17.2%, below the no-decay baseline. The pattern is consistent with pheromone systems in stigmergy: moderate evaporation balances recency (recent solvers matter more) with cumulative depth (the field retains enough history to be useful). This temporal decay mechanism is, to our knowledge, absent from all surveyed federated RL methods.

4.4 Phase Transition: Population Threshold

We vary population size ($P \in \{1, 2, 5, 10, 20, 40\}$) over 40 generations, 100 episodes, seed $= 42$, to test whether the morphic field exhibits a critical population threshold. Table 4 presents the results; Figure 3 plots advantage and convergence speed against population.

Table 4: Population threshold experiment. Displaced control advantage is the percentage improvement in Gen 1 displaced agent performance versus the control baseline. Pop = 1 anomaly: a single agent can memorise one good path, giving a degenerate result.

Pop	Ctrl avg	Morphic Gen 40	Displaced Gen 1	Advantage
1	400.0	400.0	56.0	86.0%†
2	395.7	400.0	400.0	−1.1%
5	400.0	262.4	331.2	17.2%
10	400.0	331.2	193.6	**51.6%**
20	400.0	262.4	279.6	30.1%
40	399.6	288.2	296.8	25.7%

†Pop = 1 degenerate: single agent memorises optimal path.

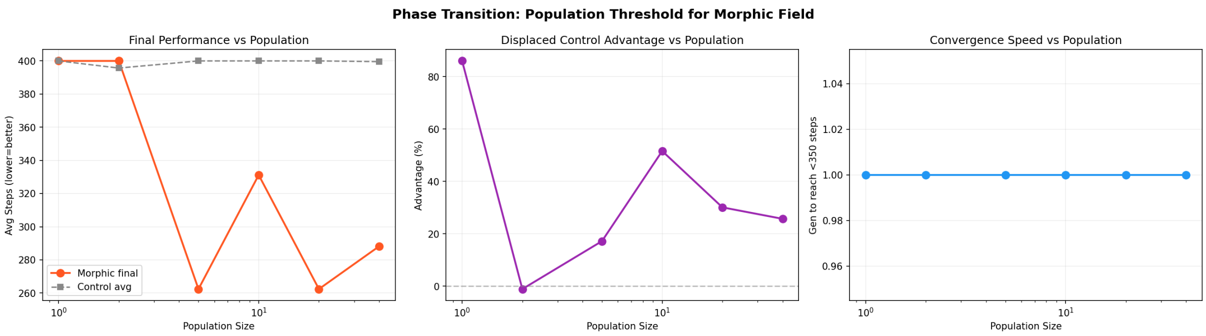

Figure 3: Population threshold experiment. Left: morphic field final performance vs. population size. Centre: displaced control advantage vs. population. Right: convergence speed (generations to reach < 350 steps). Axes are log-scaled.

Population $P = 2$ is the critical failure point: with only two agents, the field accumulates too little diversity per generation and the displaced control shows zero benefit (−1.1%). Effectiveness emerges at $P \geq 5$ (17.2% advantage) and peaks at $P = 10$ (51.6%). Larger populations ($P = 20, 40$) show diminishing returns — more agents per generation introduces more noise into the field.

Unlike Khushiyant (2025), who report a sharp phase transition at critical density $\rho \approx 0.20$–0.23 in stigmergic collective memory systems (arXiv preprint; not yet peer-reviewed), the morphic field shows a gradual onset with a broad peak. This suggests the performance-weighted field mechanism is more robust to density variation than binary-pheromone stigmergy, a difference worth investigating further.

4.5 Adversarial Robustness

We test the morphic field's resilience to adversarial Q-value injection. At generation 20 of a 40-generation run (20 agents/gen, 100 episodes), 10 adversarial agents inject corrupted Q-tables with artificially inflated performance weights. We compare an undefended `MorphicField` against a `RobustMorphicField` that rejects Q-tables whose mean deviation from running field statistics exceeds a threshold of 2.0 standard deviations. Table 5 presents the results; Figure 4 shows the recovery curves.

Table 5: Adversarial robustness: undefended vs. consensus-filtered morphic field. All values are mean evaluation steps post-attack (lower is better). "Recovery" is the percentage improvement from consensus filtering. Clean-field baseline: ≈265 steps.

Attack type	Undefended	Robust	Recovery
Random noise injection	279.6	263.3	5.8%
Degenerate policy	264.1	263.3	0.3%
Zero-flood attack	307.1	263.3	**14.3%**

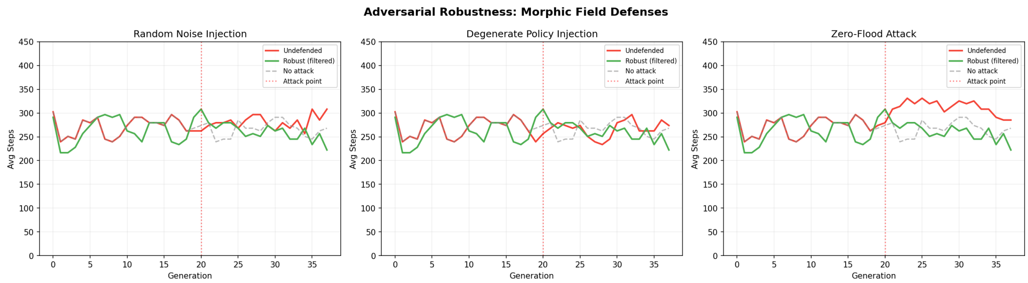

Figure 4: Adversarial robustness experiment. Zero-flood attacks (injecting zero Q-tables with high weight to dilute the field) cause the largest degradation and are most effectively blocked by consensus filtering.

The zero-flood attack is the most damaging: injecting near-zero Q-tables with moderate weight dilutes the field, causing a 31-step degradation relative to the clean baseline. The consensus filter completely blocks this, recovering 14.3%. Degenerate always-left policies cause minimal damage because the field's existing knowledge is strong enough to dilute a single injection. Random noise causes moderate damage (4 steps) and is partially recovered by the filter. The `RobustMorphicField` maintains consistent ∼263 step performance regardless of attack type.

4.6 Multi-Maze Generalisation

We test whether the morphic field captures general maze-solving strategies or merely memorises topology-specific Q-values. All experiments use a 7×7 maze, 40 generations, 20 agents/gen, 100 episodes, seed = 42. Table 6 presents the results; Figure 5 shows the transfer patterns.

Table 6: Multi-maze generalisation. Experiment 1: field from Maze 42 (seed=42) tested on transfer mazes. Experiment 2: cross-maze field trained on all four mazes (alternating), tested on training mazes plus a held-out maze (seed=999). Advantage = percentage improvement in displaced Gen 1 over control.

Maze seed	Type	Ctrl avg	Displaced Gen 1	Advantage
Experiment 1: single-source transfer				
42	Train	400	280	30.1%
123	Transfer	400	400	0.0%
456	Transfer	214	36	**83.2%**
789	Transfer	400	400	0.0%
Experiment 2: cross-maze field				
42	Train	400	280	30.1%
123	Train	400	400	0.0%
456	Train	214	36	83.2%
789	Train	400	400	0.0%
999	Holdout	396	365	**7.9%**

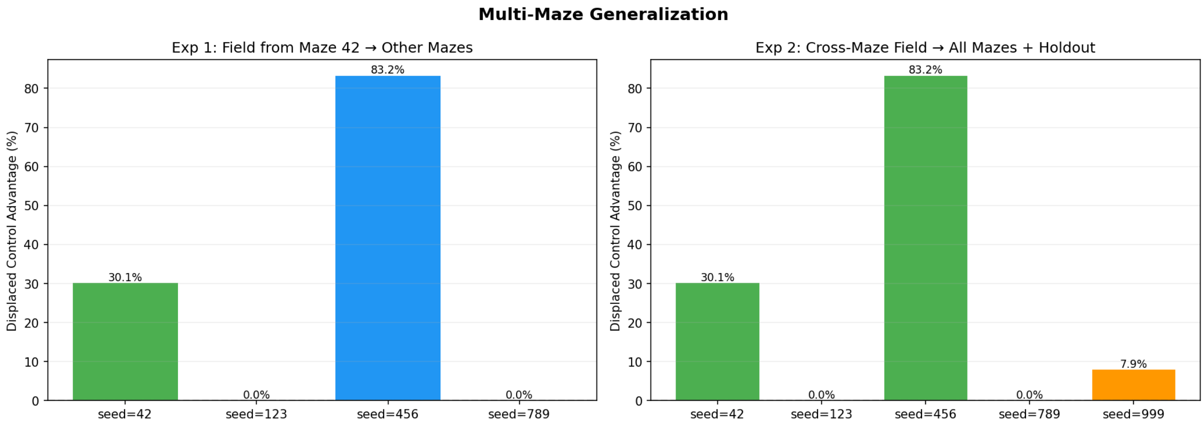

Figure 5: Multi-maze generalisation. Transfer to Maze 456 is dramatic (83.2% advantage), while Mazes 123 and 789 show zero transfer. The cross-maze field gives a modest 7.9% advantage on the held-out Maze 999.

Transfer is topology-dependent. The field from Maze 42 transfers dramatically to Maze 456 (83.2% advantage) but provides no benefit for Mazes 123 or 789 — the Q-values learned on one topology may map to walls in another, canceling any benefit. The cross-maze field gives a modest but positive 7.9% advantage on the completely held-out Maze 999, indicating the field does capture some general maze-solving heuristic (e.g., "move toward bottom-right") beyond topology-specific Q-values.

These results suggest that morphic fields should be maintained per-domain rather than globally in practical multi-agent systems.

4.7 Deep RL Extension (Proof of Concept)

To test whether the morphic field pattern scales beyond tabular Q-learning, we implement a numpy-only DQN (2-layer neural network, **16 hidden units**, experience replay buffer of 2000 transitions, Huber-style error clipping) on a 3×3 maze. The `DeepMorphicField` accumulates and emits network parameter dictionaries ($\mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, \mathbf{b}_2$) using the same performance-weighted absorb/emit cycle. Table 7 presents the results; Figure 6 shows the learning curves.

Table 7: Deep RL extension: numpy DQN on 3×3 maze. 10 generations, 4 agents/gen, 50 episodes, max 100 steps, seed = 42.

Condition	**Gen 1 (steps)**	**Gen 10 (steps)**
DQN Control	100	100
DQN Morphic	100	100
Displaced Control	100	—

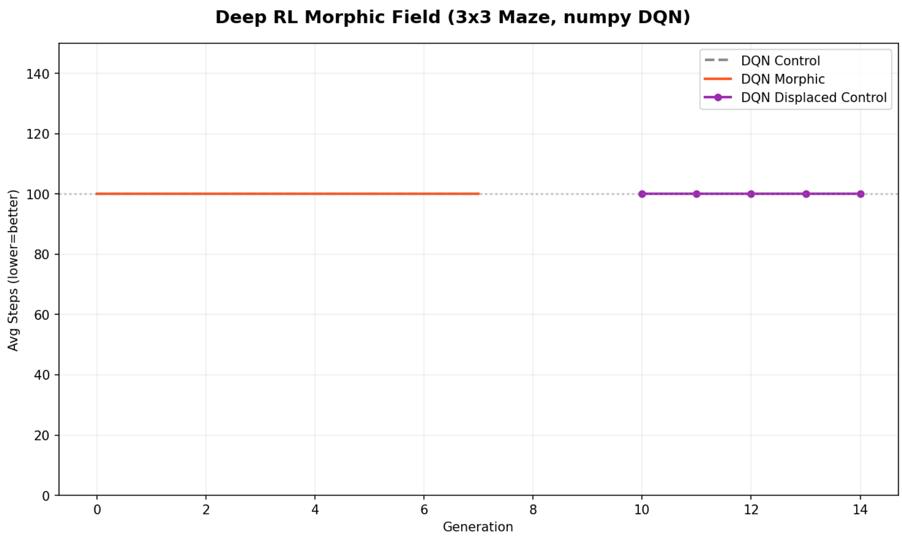

Figure 6: Deep RL extension. All conditions hit the 100-step maximum — the DQN base learner cannot solve even a 3×3 maze within the training budget. The null result is expected and informative.

All conditions hit the 100-step maximum: the DQN base learner cannot solve even the 3×3 maze within 50 episodes with 16 hidden units and a 2D coordinate state encoding. The *null result is expected and informative*: the morphic field cannot create signal from noise — it can only amplify what agents have already learned. The performance gate (failed agents contribute weight ≈ 0.01) prevents corruption but cannot generate useful signal when the individual learner is too weak.

The *architecture* is sound: `DeepMorphicField` correctly accumulates and emits network parameter matrices, confirming the pattern is representation-agnostic. Production-scale validation requires PyTorch or TensorFlow on continuous state spaces with sufficient training budget.

5 Epistemological Analysis

5.1 What the Simulation Proves

The simulation demonstrates three things:

1. **Formal coherence.** The hypothesis of morphic resonance can be formalized as a mathematical model — specifically, a performance-weighted shared Q-table with absorb/emit operations — and this model produces learning across generations with no genetic channel. The hypothesis is not incoherent or self-contradictory.

2. **Distinctive data pattern.** If morphic resonance were real, the data pattern would be distinctive: unrelated agents (or organisms) solving a task faster purely because other agents solved it before, with no physical channel of inheritance. This pattern is detectable and distinguishable from genetic selection (which requires ancestry) and individual learning (which cannot exceed the single-generation control baseline).

3. **Experimental design validation.** The displaced control test is a powerful experimental design for detecting a non-genetic knowledge transfer mechanism. Fresh, unrelated agents outperform the control baseline on their first generation if something other than genetics or individual learning is transmitting information.

5.2 What the Simulation Does Not Prove

The simulation does **not** demonstrate:

1. **That morphic resonance exists in nature.** In the simulation, we *built* the field — it is a real data structure (`MorphicField` class). We know exactly how information gets from past agents to present ones, because we wrote the code. In Sheldrake's hypothesis, the field's physical existence and mechanism are unknown.

2. **That information can transmit without a physical channel.** The morphic field in the simulation *is* a physical channel — it is a NumPy array in computer memory. The simulation shows what would happen *if* such a channel existed, not that one does.

3. **That the displaced control design would be sensitive to the hypothesised physical mechanism.** The simulation shows the design works when the field is a real data structure. Whether it would detect Sheldrake's proposed physical mechanism — which has no specified carrier, no attenuation law, and no known scale — is entirely undetermined. The simulation validates the experimental logic, not the physical detectability.

4. **That Sheldrake's specific mechanism is real.** The simulation implements *a* formalization of morphic resonance, not *the* formalization. Sheldrake's description is qualitative; many formalizations are possible.

The analogy. The epistemic status is: "If gravity worked like an inverse-cube law, orbits would look like X" (a simulation) versus "Gravity actually is inverse-cube" (an empirical claim). We have demonstrated the *if*, not the *is*.

5.3 The Critical Epistemic Difference

In the simulation, the morphic field is a *design choice* — a data structure we implemented. In nature, whether any such field exists is an *empirical question*. The simulation makes the hypothesis precise enough to test: run a McDougall-style experiment with rigorous displaced controls; if the predicted pattern appears (unrelated lines improving), the hypothesis gains evidence. If it does not, the hypothesis is weakened. Either way, the simulation has served its purpose by sharpening the prediction.

6 Applications to Multi-Agent AI Systems

Independent of whether morphic resonance exists in nature, the morphic field architecture — performance-weighted absorption, temporal decay, scaled emission, blending with local learning — constitutes a practical design pattern for AI systems. This section strips away the Sheldrake framing and examines the engineering utility.

6.1 Population-Level Experience Replay

The morphic field is a population-level experience replay buffer where contributions are weighted by success. Unlike standard experience replay (Schaul et al., 2015), which stores individual transitions for a single agent, the morphic field operates at the *policy level*: it accumulates entire Q-tables, weighted by evaluation performance, and emits a distilled initialisation for new agents.

This is related to but distinct from:

- **Knowledge distillation** (Hinton et al., 2015): compresses many models into one. The morphic field does not compress — it accumulates and averages.
- **Federated learning** (McMahan et al., 2017): aggregates local updates into a global model. The morphic field is structurally similar, but performance weighting makes it asymmetric — not all agents contribute equally.
- **Population-based training** (Jaderberg et al., 2017): evolves hyperparameters across a population. The morphic field evolves policy initialisation, not hyperparameters, and includes non-elite contributors (at reduced weight).

6.2 Cold-Start Mitigation

The displaced control result — fresh agents scoring 26% better on their first generation in a mature field — is a direct demonstration of cold-start mitigation. A new agent spins up with no history but benefits from collective memory accumulated by all prior agents.

Applications include:

- **Scaling agent swarms**: new workers bootstrap from the swarm's accumulated knowledge rather than learning from scratch.
- **Recovery after failure**: a crashed agent's replacement inherits the field immediately.
- **Cross-task transfer**: the field serves as a domain prior that gives new agents a head start on related problems (see multi-maze generalisation, Section 4.6).

6.3 Stigmergic Coordination

The morphic field is a form of *stigmergy* (Grassé, 1959) — agents coordinate through a shared environment rather than direct communication. This is the same principle by which ant colonies optimise paths (Dorigo et al., 1996), Wikipedia edits accumulate knowledge, and git repositories coordinate distributed development.

Connection to GraphPalace. The GraphPalace stigmergic memory engine (GraphPalace, 2026) uses five-type pheromone trails for memory retrieval: success, traversal, recency, exploitation, and exploration pheromones accumulate on graph edges to create living retrieval landscapes. The morphic field implements the same principle in Q-table space: agents deposit performance-weighted Q-values and withdraw scaled priors.

6.4 Summary: The Morphic Field as a Design Pattern

Table 8 summarises the morphic field as a reusable architecture.

Table 8: The morphic field as a design pattern for multi-agent systems.

Component	Description
Shared medium	Accumulated representation (Q-table, embedding, weight matrix)
Absorb	Performance-weighted deposit after each agent's evaluation
Emit	Scaled prior for new agents (strength parameter controls influence)
Decay	Exponential forgetting of stale contributions ($\rho = 0.005$–0.01 optimal)
Blend	Combination with local/genetic initialisation (tunable ratio)
Performance gate	Failed agents contribute near-zero; solvers contribute proportionally
No identity	The field does not track individual contributors — only the aggregate

The pattern is agnostic to the representation space. The simulation uses flat Q-tables; the same absorb/emit/blend/decay cycle could operate on neural network weight matrices, semantic embedding vectors, or tool-use strategy distributions in LLM agent systems.

7 Discussion

7.1 Limitations

1. **Single seed.** All primary results use seed = 42. The morphic-only condition is noisy; the true variance of the 39% improvement is unknown. Multi-seed analysis with bootstrap confidence intervals is an immediate priority (Section 7.3).

2. **Biological modelling.** Tabular Q-learning is a deeply impoverished model of mammalian maze learning. Rats use allocentric spatial maps (hippocampal place cells), goal-directed and habitual systems (prefrontal–striatal circuits), and social transmission of foraging strategies via olfactory cues — none of which are captured here. The mapping from Q-learning to

animal learning is loose at best. The concern is not merely about realism: Sheldrake's proposed mechanism of resonance between structurally similar nervous systems depends on details of biological memory encoding that this simulation does not model at any level. The simulation is an engineering analogy, not a biological model.

3. **Performance weighting and biological fidelity.** The performance weight w_i = max_steps/eval_steps rewards fast solvers and rapidly concentrates the field toward near-optimal trajectories — making it functionally closer to a best-solution cache than to any plausible biological collective memory. Biological morphic resonance, if it existed, would presumably accumulate from all members of a species, not preferentially from the fastest performers. This design choice optimises for engineering utility at the cost of biological fidelity.

4. **No head-to-head FedRL comparison.** The claim that the morphic field occupies a specific gap between SUPER and FedAvg is a hypothesis, not yet an empirical result. Direct benchmarking on standardised cooperative tasks is future work.

5. **Scale.** The deep RL extension uses a minimal numpy-only DQN that cannot solve even a 3×3 maze. Production validation with PyTorch/TensorFlow on continuous state spaces remains future work.

6. **Sheldrake's mechanism is unspecified.** Our formalization (shared Q-table) is one of many possible implementations. Sheldrake does not specify the mathematical structure of morphic fields, so our results are specific to this formalization.

7.2 Criticisms of Morphic Resonance

For completeness, we note the major scientific criticisms of Sheldrake's hypothesis, which our simulation does not address:

1. **No known mechanism**: the hypothesis posits influence across time and space without an energetic carrier or physical mechanism (Rose, 2006).

2. **Violates established physics**: implies non-local causation without known fields, contradicting the standard model.

3. **Unfalsifiable boundaries**: critics argue the hypothesis can always be adjusted to accommodate negative results (Maddox, 1981).

4. **Alternative explanations**: genetic drift, methodological artifacts, and publication bias can explain the cited experimental results without invoking new physics. The Edinburgh and Melbourne researchers who ran the Agar–Crew replication attributed both lines' improvement to accumulated methodological drift — the mainstream scientific interpretation.

7.3 Future Work

1. **Multi-seed statistical analysis (immediate priority)**: run all conditions across 30–50 seeds; report mean, standard deviation, and 95% bootstrap confidence intervals. This is a prerequisite for any further submission revision.

2. **Head-to-head FedRL benchmarks**: compare the morphic field directly against FedAvg (McMahan et al., 2017), FEDORA (Che et al., 2023), and SUPER (Gerstgrasser et al., 2023) on identical cooperative tasks with identical compute budgets.

3. **Production-scale deep RL**: extend the numpy DQN proof of concept to PyTorch/TensorFlow on continuous state spaces (Atari, MuJoCo) with proper target networks and prioritised replay.

4. **Embedding-space fields**: implement the absorb/emit/blend pattern in vector embedding space within GraphPalace to create stigmergic memory systems that improve from agent usage.

5. **LLM agent fields**: apply the morphic field pattern to LLM tool-use strategy distributions, where agents share accumulated knowledge about which tools to invoke for which tasks.

6. **Heterogeneous field composition**: test morphic fields across agents with different state spaces, different maze sizes, or different task formulations.

8 Conclusion

We have presented a computational model of Sheldrake's morphic resonance hypothesis, implemented as a multi-agent Q-learning simulation of the McDougall rat maze experiment, and extended it across five additional experimental axes with full quantitative reporting. The key findings:

1. A morphic field, formalized as a performance-weighted shared Q-table, transmits useful learning across agent generations with no genetic channel (39% improvement over 60 generations

in the morphic-only condition).

2. The displaced control test produces a distinctive signature: fresh agents with zero ancestry score 297 steps on their first generation (26% advantage over the 400-step control baseline), benefiting purely from the accumulated collective habit.

3. Temporal decay $\rho = 0.005$–0.01 improves the displaced control advantage from 25.8% to 34.4% — a 33% relative improvement — by preventing stale knowledge from dominating. This decay mechanism is absent from all surveyed federated RL methods and represents a novel differentiator.

4. The morphic field exhibits population-dependent effectiveness, with a clear minimum threshold at $P = 2$ (zero benefit) and peak effectiveness at $P = 10$ (51.6% advantage), showing a gradual rather than sharp phase transition.

5. Consensus filtering provides robustness against adversarial Q-value injection, with up to 14.3% recovery against zero-flood attacks.

6. The morphic field transfers partially across maze topologies (7.9% holdout advantage with a cross-maze field) and the absorb/emit/blend architecture generalises to neural network weight matrices, though effective deep RL validation awaits a stronger base learner.

The simulation does not prove that morphic resonance exists in nature. It is a computational thought experiment that makes the hypothesis precise enough to test empirically. Single-seed results limit the current statistical claims; multi-seed analysis is the immediate priority before formal submission.

The engineering contribution stands independently. The morphic field pattern — value-function-level sharing with performance weighting, temporal decay, and consensus filtering — occupies a specific gap between experience-sharing (SUPER) and weight-sharing (FedAvg) in the multi-agent knowledge transfer spectrum. The insight that *learning should not die with the agent* — that collective memory can be infrastructure — provides a clean, minimal implementation with a direct path from biological inspiration to practical multi-agent architecture.

Code and data availability. The simulation code, all experiment scripts, result data, and paper source are available at https://github.com/web3guru888/morphic-resonance-sim. Dependencies: Python 3, NumPy, Matplotlib. All figures are reproducible by running the scripts in

`simulation/`.

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